

Development of austenite free ADI (austempered ductile cast iron)

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Abstract

The primary goal of this investigation was to create austempered ductile cast iron (ADI) with a fully ferritic microstructure without compromising its mechanical properties. This was achieved by applying a novel heat treatment process. This process consists of austempering and subsequent isothermal tempering below the inter critical (A_1) temperature of ductile cast iron. Ductile cast iron samples were initially austenitized at 927 °C (1700 °F) for 2 h and then austempered at three different temperatures 260 °C (500 °F), 316 °C (600 °F) and 385 °C (725 °F) and finally isothermally tempered at 484 °C (900 °F) for 2 h. This resulted in a fully ferritic microstructure. The effect of this tempering on the physical and mechanical properties of the material was examined and compared with conventionally processed ADI. Tests results show that when ADI is austempered at 260 °C (500 °F) and subsequently tempered at 484 °C (900 °F) it has significantly better mechanical properties than the samples initially austempered at other two temperatures (i.e. 316 °C (600 °F) and 385 °C (725 °F)).

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1. Introduction

Austempered ductile cast iron (ADI) is an alloyed and heat treated nodular cast iron. It has emerged as a major engineering material in recent years because of its attractive properties such as good ductility and high strength [1–3], good wear resistance [4], high fatigue strength [5–9] and fracture toughness [10–14]. Because of these unique properties, ADI is now used extensively in many structural applications in automotive industry, defense and earth moving machineries among others.

The attractive properties of ADI are related to its unique microstructure. When ductile cast iron is austempered, it produces a microstructure that consists of ferrite (α) and high carbon austenite (γ_{HC}) with graphite nodules dispersed in it. This is different from the austempered steels where the microstructure consists of ferrite and carbide (bainite). Because of this difference, the product of austempering reaction in ductile iron is often referred to “ausferrite” rather than bainite as in steels. Large amount of silicon present in ductile iron suppresses the precipitation of carbide during the austempering reaction and retains substantial amount of stable high carbon austenite (γ_{HC}). Small

amounts of alloying elements such as nickel, molybdenum and copper are generally added to ADI so that it has sufficient hardenability to be quenched to the austempering temperature without forming pearlite.

The starting material for the development of ADI is ductile or nodular cast iron. It is then subjected to an isothermal heat treatment process known as austempering. The austempering process involves a first austenitizing the casting in the temperature range of 871–982 °C for sufficient time to get a fully austenite (γ) matrix and then quenching it to an intermediate temperature range of 260–400 °C. The casting is maintained at this temperature for 1–4 h. During the austempering process a two stage phase transformation reaction takes place in ADI. In the first reaction, the austenite (γ) decomposes into ferrite and high carbon austenite (γ_{HC}). If the casting is held at the austempering temperature for too long, a second reaction occurs which causes further decomposition of high carbon austenite (γ_{HC}) into ferrite (α) and carbide. The microstructure in this case contains ϵ carbide which makes the material brittle [15,16]. Therefore, this second reaction must be avoided during the austempering process. Thus the best combination of mechanical properties is obtained in ADI, after the completion of the first reaction but before the onset of the second reaction. The time period between the completion of the first reaction and the onset of second reaction is termed as the “process window”. One important function

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of the alloying elements added to ADI is to enlarge [2,3,16] this process window.

While ADI has many attractive mechanical properties, it has some limitations. The ADI microstructure consists of ferrite and austenite. Austenite has a higher work hardening rate than ferrite [17–19] and this causes problems in machining of ADI. The traditional bainitic austenite structure in ADI creates machinability and tool life problems because austenite is mechanically unstable and transforms to martensite during machining. Since most of the ADI components in automotive, aircraft and other manufacturing industries need to be machined, a fully ferritic structured ADI (austenite free) could potentially reduce machining cost of these components. The thermal shock resistance as well as thermal fatigue resistance depends on the thermal conductivity of the materials. FCC (face centered cubic) Austenite has a lower [20] thermal conductivity than bcc (body centered cubic) ferrite, thus ADI has a relatively low thermal shock and thermal fatigue resistance because of the mixed microstructure consisting of ferrite and austenite. For many applications structural components require high thermal shock resistance. As for example, brake rotors in automobiles and trucks need excellent thermal shock resistance which in turn requires very high thermal conductivity. Another problem of austenite is that it has much higher [20,21] co-efficient of thermal expansion than ferrite which can cause problems in applications where thermal fatigue is important.

Thus if a fully ferritic structured ADI can be developed without compromising its mechanical properties it will have significant advantages and have many potential applications. The primary objective of this investigation was thus to create ADI with a fully ferritic (austenite free) microstructure. The secondary objective was to evaluate the mechanical, electrical and thermal properties of this ADI to establish its suitability for potential applications where thermal shock and thermal fatigue are important.

2. Experimental procedure

2.1. Material

The material used in the present investigation is an alloyed nodular cast iron of conventional composition. The chemical composition of the material is reported in Table 1. The microstructure of the as cast material is shown in Fig. 1. The



Fig. 1. Microstructure of As cast material.

as cast material was predominantly pearlitic (over 80%) in nature. The material was cast in the form of round cylindrical bars of 2.54 mm (1 in.) diameter and 370 mm (14 in.) long and keel blocks. For tensile testing, cylindrical samples were prepared as per ASTM standard E-8 [22] from the round bars. For fracture toughness test, compact tension samples were prepared as per ASTM standard E-399 [23] from these cast keel blocks. The compact tension specimens had a thickness $B=20.2$ mm and width $W=40.4$ mm. The aspect ratio a/W was kept at $a/W=0.45$. The details of these specimens have been reported elsewhere [24–26].

2.2. Heat treatment

Two batches of samples were prepared. One batch of samples was processed by conventional austempering process. These samples were initially austenitized at 927 °C (1700 °F) for 2 h and then austempered for 2 h at three different temperatures: 260 °C (500 °F), 316 °C (600 °F) and 385 °C (725 °F). The 2 h time is within the process window of ADI. The second batch of samples was initially processed in the above manner (austenitized at 927 °C (1700 °F) and austempered at the above three temperatures). Afterwards these samples were further tempered at 484 °C (900 °F) for 2 h. An initial investigation by these authors has shown that after tempering at 484 °C (900 °F) for 2 h, a fully ferritic structure can be obtained in ADI. Finally these samples were air cooled to room temperature.

2.3. Tensile testing

Tensile testing of all these samples (after heat treatment) was carried out as per ASTM standard E-8 [22]. The tests were carried at a constant engineering strain rate of $4 \times 10^{-4} \text{ s}^{-1}$ on a MTS (material test system) servo hydraulic machine at room temperature and ambient atmosphere. Load and displacement plots were obtained on a X–Y recorder and from these load displacement diagrams yield strength, ultimate tensile strength and % elongation values were calculated. The average values from four test samples are reported in this paper.

Table 1
Chemical composition of the material (wt.%)

Element	Percentage (wt.%)
C	3.45
Si	2.48
Mn	0.4
Ni	1.5
Mo	0.5
Cu	0.3
Mg	0.05
S	0.012
P	0.013
Fe	Remaining

2.4. Fracture toughness testing

After heat treatment, the compact tension samples were ground and then polished with 600 grit emery paper. The specimens were then pre cracked in fatigue at a ΔK level of 10 MPa $\sqrt{\text{m}}$ with a load ratio $R=0.10$ to produce 2 mm long sharp crack front in accordance with ASTM standard E-399 [23]. After fatigue cracking the specimens were loaded in tension in a servo hydraulic MTS (material test system) test machine and the load displacement diagrams were obtained with a clip gauge in the knife edge attachment on the specimens. From these load displacement diagrams, P_Q values were determined using the 5% secant deviation technique as per ASTM standard E-399 [23]. From these P_Q values, K_Q values were determined using the standard stress intensity factor calibration function for the compact tension specimens. Since these K_Q values satisfied all the requirements of a valid K_{IC} test as per ASTM standard E-399, they were all valid K_{IC} values. Four identical test specimens were tested from each heat treated condition. Values reported in this paper are averages from these four tests.

2.5. Metallography and X-ray diffraction

Microstructures of all the samples were examined by optical microscopy after polishing and etching with 2% nital solution. X-ray diffraction (XRD) analysis was performed to estimate the austenite content and the percentage of carbon in austenite following the procedure of Rundman and Klug [27]. XRD was done using a monochromatic copper $K\alpha$ radiation at 40 kV and 100 mA. A Rigaku rotating head anode diffractometer was used to scan the angular 2θ range from 42° to 48° at a scanning speed of $0.25^\circ/\text{min}$, and $72\text{--}92^\circ$ at a scanning speed of $1^\circ/\text{min}$. The profiles were analyzed using Jade 5 software to obtain the peak positions and the integrated intensities of $\{111\}$, $\{220\}$ and $\{311\}$ planes of fcc (face centered cubic) austenite and $\{110\}$ and $\{211\}$ planes of bcc (body centered cubic) ferrite. The volume fraction of ferrite (X_α) and austenite (X_γ) were determined by the direct comparison method using the integrated intensities of the above planes [28].

The carbon content of the austenite was determined by the equation [29]:

$$a_\gamma = 0.3548 + 0.0044 C_\gamma$$

where a_γ is the lattice parameter of austenite in nanometer and C_γ is the carbon content of austenite in wt.%. The $\{111\}$, $\{220\}$ and $\{311\}$ planes of austenite were used to estimate the lattice parameter. Four samples were examined from each heat treated condition and the data reported here is the average from four samples.

3. Results and discussions

3.1. Microstructure (non-tempered samples)

The microstructures of the samples processed by conventional austempering process (before final tempering at 900°F) are shown in Fig. 2(a–c) for the three different temperatures.

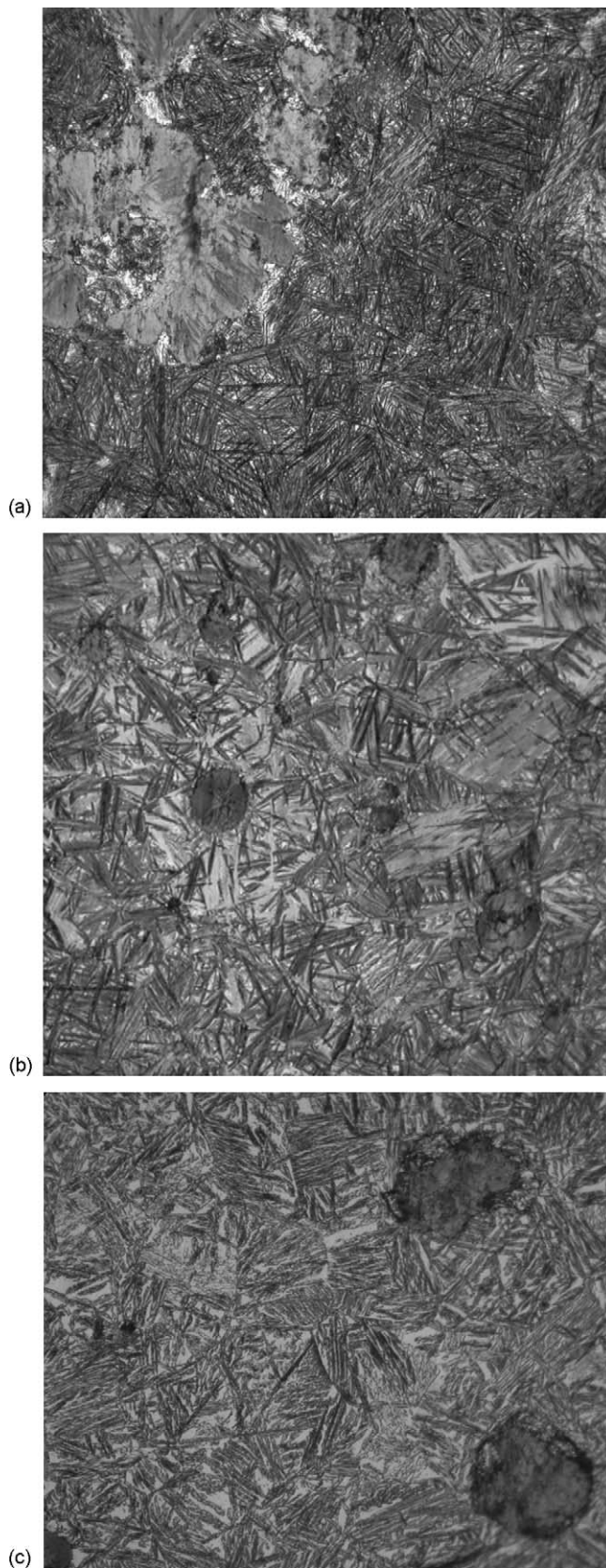


Fig. 2. (a) Microstructure of samples after austempering at 500°F . (b) Microstructure of samples after austempering at 600°F . (c) Microstructure of samples after austempering at 725°F .

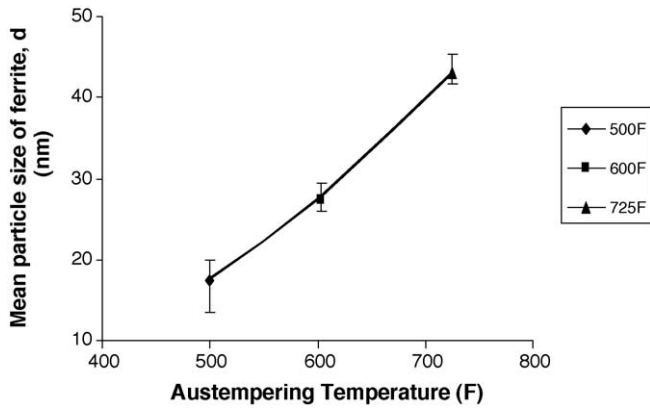


Fig. 3. Effect of austempering temperature on ferrite cell size.

The austempered microstructure shows a matrix of dark needle shaped ferrite plus a bright etching austenite with graphite nodules dispersed in it. The austenite and ferrite progressively coarsen as the austempering temperature increases from 260 °C (500 °F) to 385 °C (725 °F). At lower austempering temperature (260 °C) very fine ferrite and austenite was observed. On the other hand, at higher austempering temperature (385 °C) coarse and feathery ferrite characteristic of upper baintic structure was observed. An estimate of the ferritic cell size (d) was obtained from the X-ray diffraction profile from the breadth of $\{211\}$ diffractometer peaks of ferrite using the Scherrer equation [28]:

$$d = \frac{0.9\lambda}{\beta \cos \theta}$$

where λ is the wave length, β the breadth of the $\{211\}$ peak of ferrite at half height in radians and θ is the Bragg angle. In Fig. 3, the ferritic cell size has been plotted against the austempering temperature. It shows that the ferritic cell size (d) increases as the austempering temperature increases confirming the coarsening of the ferrite with an increase in austempering temperature.

Figs. 4 and 5 report the influence of austempering temperature on the volume fraction of ferrite and austenite, respectively. As the austempering temperature increases the volume fraction of ferrite decreases and correspondingly the volume fraction of

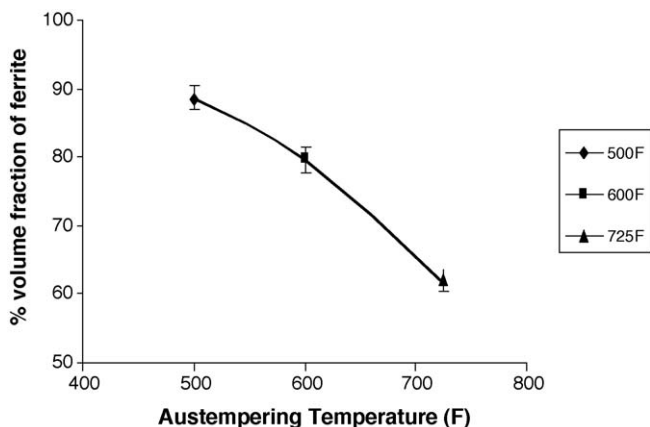


Fig. 4. Effect of austempering temperature on volume fraction of ferrite.

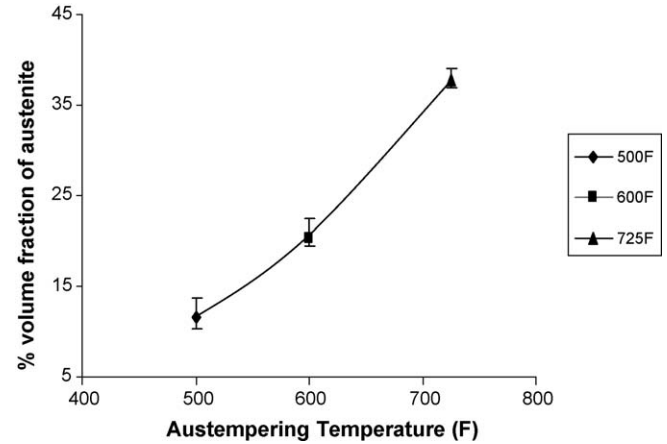


Fig. 5. Effect of austempering temperature on volume fraction of austenite.

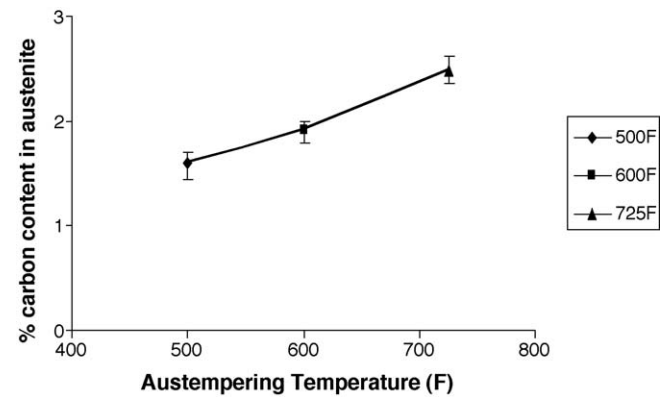


Fig. 6. Effect of austempering temperature on carbon content of austenite.

austenite increases. The reason for this has already been discussed in the previous publications [2,11,12]. Fig. 6 is a plot of carbon content of austenite as a function of austempering temperature. It is evident that as the austempering temperature increases the carbon content of austenite also increases. During austempering process ferrite needles nucleate out of austenite. As the austempering proceeds, these ferrite needles grow and consequently the remaining austenite continues to absorb carbon and the carbon content of the austenite thus increases. In Fig. 7, the effect of austempering temperature on the austenite carbon

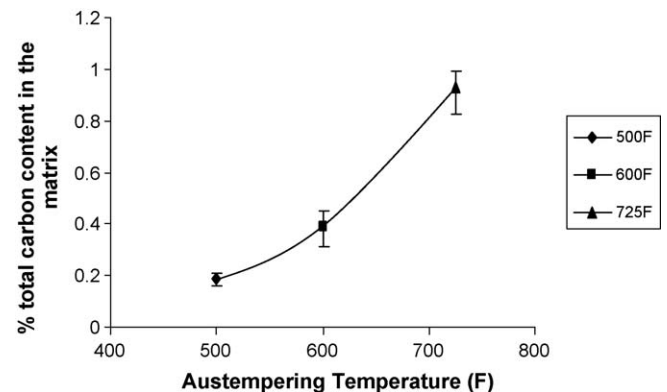
Fig. 7. Effect of austempering temperature on austenitic carbon ($X_{\gamma}C_{\gamma}$).

Table 2
Ferritic cell size of tempered and non-tempered samples

Heat treatment	Ferritic cell size, d (nm)
1700 °F/2 h–500 °F/2 h	
Before tempering	17.38 ± 3.3
After tempering	14.54 ± 1.8
1700 °F/2 h–600 °F/2 h	
Before tempering	27.35 ± 1.7
After tempering	21.09 ± 2.2
1700 °F/2 h–725 °F/2 h	
Before tempering	43.12 ± 1.8
After tempering	22.34 ± 2.6

($X_\gamma C_\gamma$) is reported. The austenitic carbon is a measure of the total carbon content of austenite and is a product of volume fraction of austenite (X_γ) and its carbon content (C_γ). It is evident from this figure that as the austempering temperature increases the austenite carbon content ($X_\gamma C_\gamma$) also increases with increase in temperature. Since both the volume fraction of austenite as well as the carbon content increases, with increase in temperature the product ($X_\gamma C_\gamma$) also increases with the increase in austempering temperature.

3.1.1. Microstructure of tempered samples

Fig. 8(a–c) report the microstructure of the austempered samples after additional tempering at 900 °F. All these samples show a totally ferritic microstructure. The X-ray diffraction profile of 725 °F austempered samples before and after tempering at 900 °F are shown in Figs. 9 and 10. While non-tempered samples show both ferrite and austenite peaks, the tempered samples show only ferrite planes i.e. after tempering at 900 °F for 2 h, in the angular 2θ range from 42° to 46° consists of only ferrite (α) (1 1 0) planes. Similarly in the angular 2θ range from 72° to 92° it consists of only one plane {ferrite α 1 (2 1 1)} plane. This indicates that after tempering there is only ferrite present in these samples and austenite content is below the resolution of the X-ray. Similar behavior was also observed from the X-ray diffraction profiles of the 500 and 600 °F austempered materials after tempering at 900 °F.

Table 2 reports the ferritic cell size of the tempered samples. It is evident from this table ferritic cell sizes are much finer at 500 °F austempered samples compared to 600 and 725 °F samples. At higher temperature, decomposition of austenite to ferrite occurs because of the difference in free energy. Austenite is in a metastable state, ferrite and graphite are stable phases in ADI. With higher temperature this metastable austenite decomposes by diffusion of carbon through grain boundaries to the existing graphite nodules. This generally occurs in unalloyed ductile iron at 750 °F. This process is similar to second reaction (as mentioned earlier) but accelerated due to higher temperature.

3.2. Mechanical properties

3.2.1. Hardness

The influence of austempering temperature on hardness of both tempered and non-tempered austempered ductile cast iron

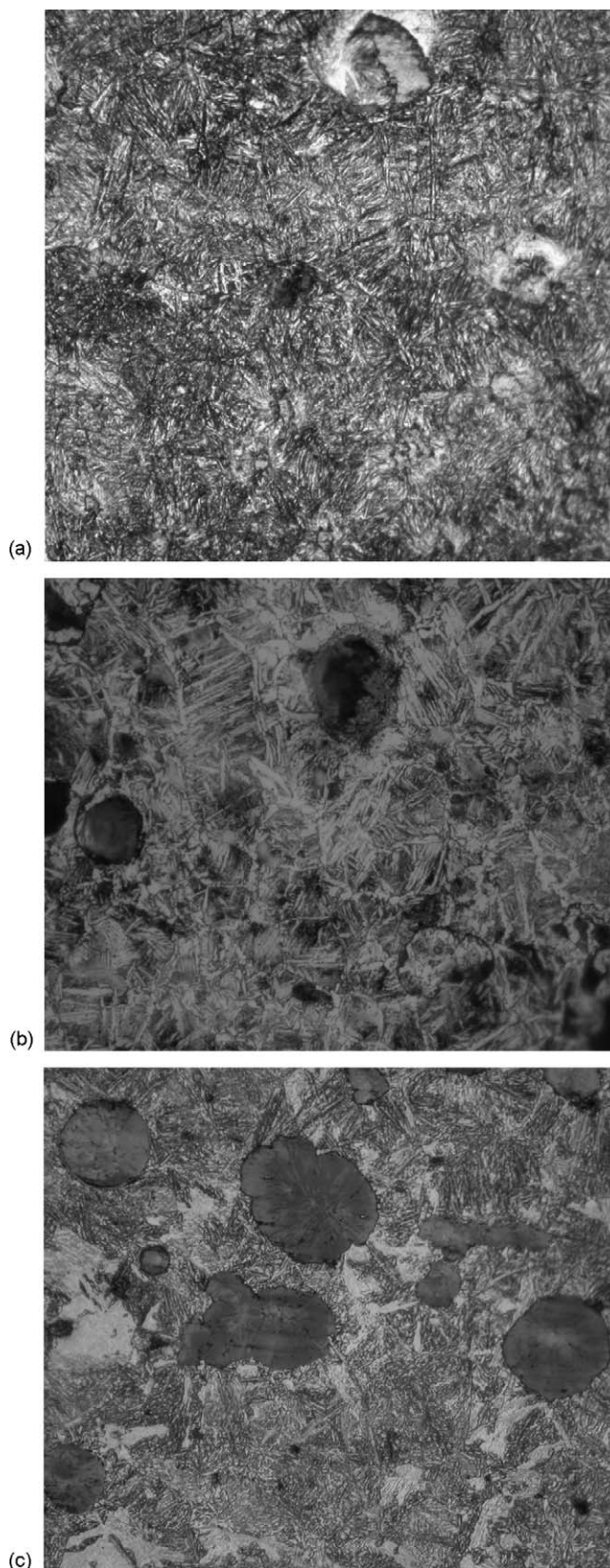


Fig. 8. (a) Microstructure of 500 °F austempered sample after tempering at 900 °F. (b) Microstructure of 600 °F austempered sample after tempering at 900 °F.

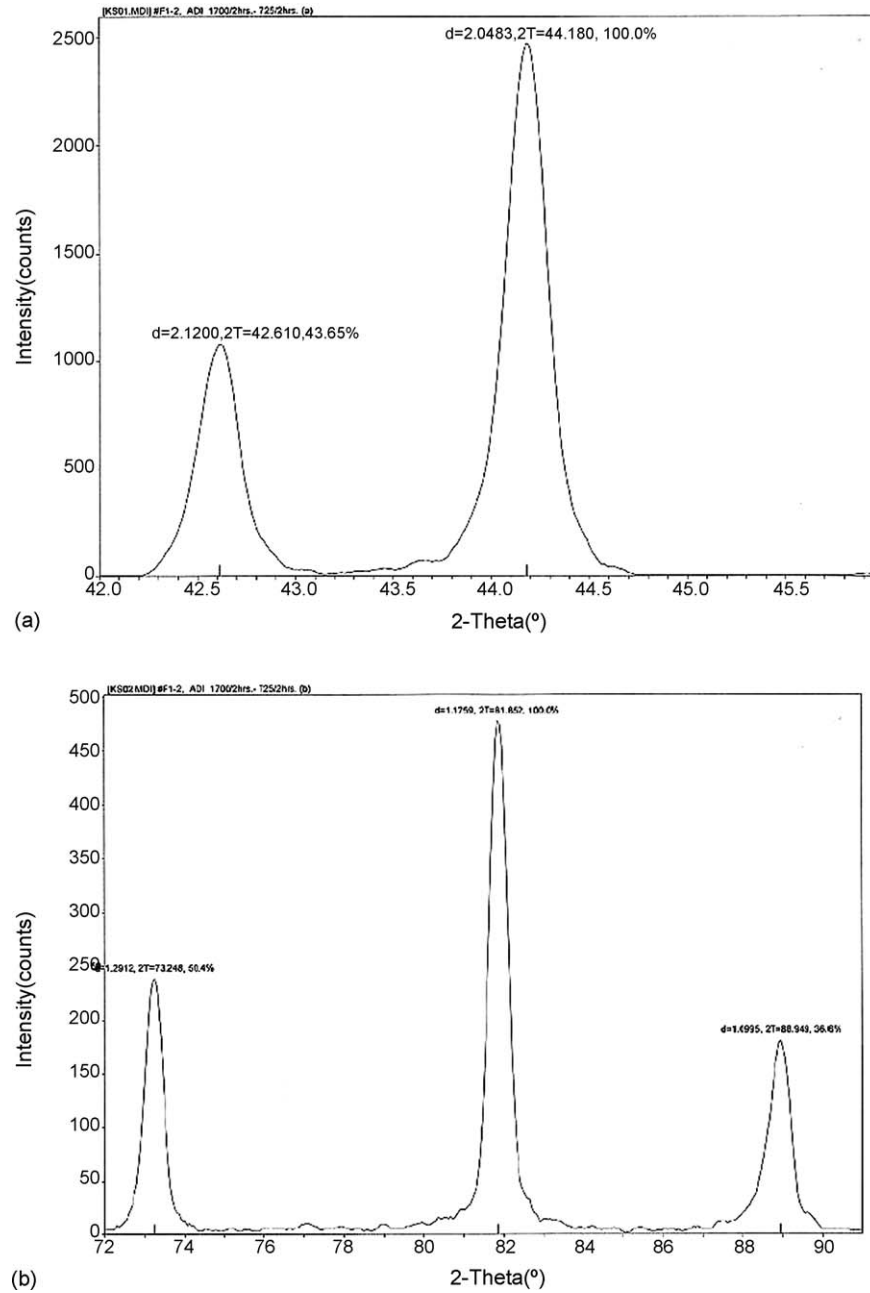


Fig. 9. (a) X-ray profile of austempered (725°) samples, before tempering (2θ range from 72° to 92°). (b) X-ray profile of austempered (725°), before tempering (2θ range from 42° to 46°).

is reported in Table 3. It is evident from this table, that there is a gradual decrease in hardness in the non-tempered samples as the austempering temperature is increased. The reason for this behavior is due to the increase in austenite content as well as coarsening of the microstructure with increase in austempering temperature. Austenite is a softer phase therefore as the austenite content increases the strength decreases and consequently hardness also decreases. Moreover the grain coarsening also causes reduction in hardness and strength [21,30,31]. In the non-tempered samples as the austempering temperature was increased austenite volume fraction increased and at the same time coarsening of both austenite and ferrite took place and this caused the reduction in hardness in these samples.

Table 3
Hardness of the material tempered and non-tempered samples

Heat treatment	Rockwell hardness on C scale (HRC)
1700 °F/2 h–500 °F/2 h	
Before tempering	46.9 ± 0.15
After tempering	39.1 ± 0.6
1700 °F/2 h–600 °F/2 h	
Before tempering	41.2 ± 0.3
After tempering	36.9 ± 0.1
1700 °F/2 h–725 °F/2 h	
Before tempering	30.1 ± 0.4
After tempering	35.0 ± 1.3

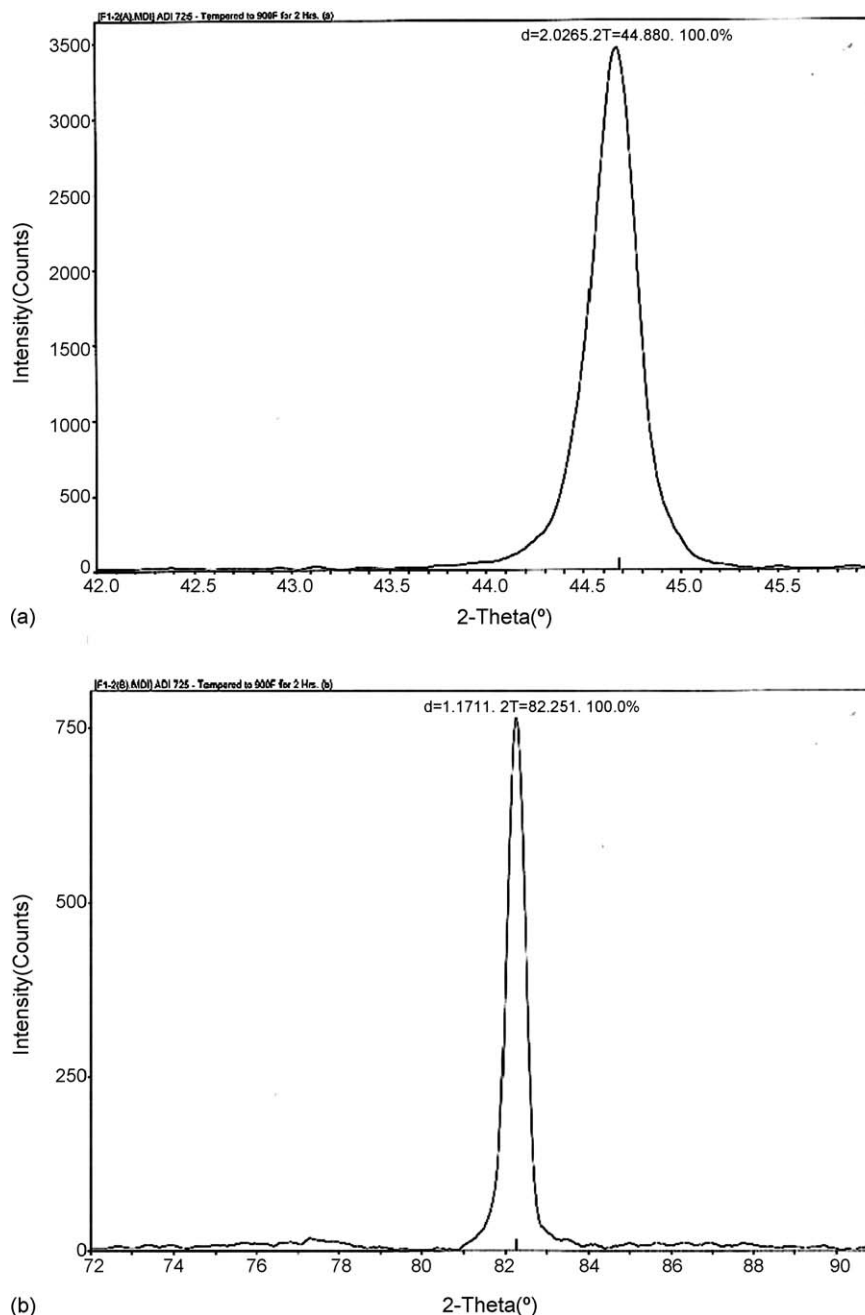


Fig. 10. (a) X-ray profile of austempered (725°) and finally tempered sample (2 θ range from 42° to 46°). (b) X-ray profile of austempered (725°) and finally tempered sample (2 θ range from 72° to 92°).

It is evident from this table that the tempered samples of 500 and 600 °F austempered samples have much lower hardness than the non-tempered samples whereas the 725 °F austempered samples have higher hardness after tempering. This higher hardness of the tempered samples of 725 °F austempered samples is probably due some precipitation of carbides in these samples.

3.2.2. Yield strength, tensile strength and ductility

Table 4 reports the influence of austempering temperature on the yield strength, tensile strength, and the ductility of both the tempered and non-tempered samples. As the austempering temperature was increased from 260 °C (500 °F) to 385 °C (725 °F),

there was a decrease in the yield and tensile strength in the non-tempered samples. This is because of the coarsening of both the austenite and ferrite with the increase in austempering temperature. It is well known (Hall–Petch equation) that the finer grains results in higher yield and tensile strengths. From Table 4, it is also evident that in the non-tempered samples, the ductility of the material increases as the austempering temperature is increased. This is due to the fact, that more of the softer and ductile austenite phase is present in the matrix at higher austempering temperatures. FCC austenite has more slip systems than bcc ferrite phase. Thus when more austenite is present in the material it has more ductility.

Table 4
Mechanical properties of the material tempered and non-tempered samples

Heat treatment	UTS (MPa)	Yield strength (MPa)	Elongation (%)	Reduction in area (%)
1700 °F/2 h–500 °F/2 h				
Before tempering	1446.4 ± 197.3	1126.2 ± 13.5	2.6 ± 1.1	2.9
After tempering	1333.9 ± 47.1	1165.8 ± 33.7	3.4 ± 1.9	3.9
1700 °F/2 h–600 °F/2 h				
Before tempering	1264.1 ± 78.0	1031.3 ± 13.9	4.4 ± 1.2	5.8
After tempering	1105.1 ± 8.9	1033.8 ± 1.7	0.9 ± 0.2	1.5
1700 °F/2 h–725 °F/2 h				
Before tempering	1069.1 ± 32.2	776.1 ± 91.8	13.7 ± 3.9	12.2
After tempering	1104.2 ± 18.9	868.1 ± 35.0	3.4 ± 0.8	4.4

Table 4 also reports the mechanical properties of the tempered samples. When compared with the non-tempered samples, there was no significant difference in yield and tensile strengths of these materials. However there was a significant decrease in ductility in tempered samples of 316 °C (600 °F) and 385 °C (725 °F) austempered material. On the other hand, for the 260 °C (500 °F) austempered and tempered samples there was no decrease in ductility. It appears that the finer scale microstructure of this material helped to retain the ductility as well as high strength of the material. While no carbide precipitation was detected at 500 and 600 °F austempered samples in X-ray diffraction plots there may be some inter granular carbide precipitation in these samples. X-ray data showed traces of carbide in these samples. Carbide precipitation can cause both reductions in ductility as well as fracture toughness [32–34]. A detailed TEM analysis is being carried out to examine this aspect.

3.2.3. Fracture toughness

The influence of the austempering temperature on the fracture toughness of tempered and non-tempered samples are reported in Table 5. In the non-tempered samples the fracture toughness values increases from 44.6 to 70.4 MPa√m as the austempering temperature is increased from 260 °C (500 °F) to 385 °C (725 °F). The maximum fracture toughness is obtained in ADI at a particular temperature where the austenite formed is fully stabilized with carbon [33]. At 500 °F austempering temperature, the microstructure contains some unreacted austenite and it is not saturated with carbon and hence it has lower fracture toughness. Several workers [14,27,24] have reported that maximum fracture toughness is obtained in ADI at an intermediate

temperature where the product $\sigma_y(X_\gamma C_\gamma)^{1/2}$ is maximum. Apparently among the non-tempered samples this occurs at 725 °F. Therefore the fracture toughness is maximum at this temperature. Earlier a mathematical model was developed for ADI by one of the authors (SKP) [35] relating fracture toughness (K_{IC}^2) to the parameter, $\sigma_y(X_\gamma C_\gamma)^{1/2}$. The model implies that the square of fracture toughness (K_{IC}^2) has a linear relationship with the parameter $\sigma_y(X_\gamma C_\gamma)^{1/2}$. In Fig. 13, we have plotted (K_{IC}^2) against the parameter $\sigma_y(X_\gamma C_\gamma)^{1/2}$. It is evident that the higher the value of $\sigma_y(X_\gamma C_\gamma)^{1/2}$, the higher the fracture toughness thus validating the earlier model.

3.2.4. Tempered samples

In the case of tempered samples, it was observed that the fracture toughness values remains more or less constant in 260 °C (500 °F) austempered samples. It can be seen from Table 5, that there is a considerable decrease in fracture toughness value when ADI is tempered at 900 °F after austempering at 600 and 725 °F compared to non-tempered samples. After tempering there was some increase in ferritic cell size at these two temperatures. Furthermore, there was lack of presence of austenite. Combination of these two factors has resulted in very low fracture toughness in these samples. Therefore it appears for high fracture toughness carbon stabilized austenite is absolutely essential in ADI. It is also obvious finer grain size will also result in higher fracture toughness.

From the above discussion it becomes obvious that in 500 °F austempered samples there was no reduction in mechanical properties after tempering at 900 °F even though it has a fully ferritic structure. On the other hand at 600 and 725 °F austempered samples there is significant decrease in the ductility and fracture toughness after tempering. The reason for this is not completely clear but it appears precipitation of carbide has caused reduction in ductility as well as fracture toughness in 725 °F tempered samples.

3.2.5. Fractography

3.2.5.1. Non-tempered samples. Fractographs of the samples austempered for 2 h at different temperatures before tempering are shown in Fig. 11(a–c). A mixed mode of fracture with cleavage and few dimples were observed in the samples austempered at 600 and 725 °F. However, the ADI samples austempered at lower austempering temperature of 500 °F show fewer micro

Table 5
Fracture toughness of tempered and non-tempered samples

Heat treatment	Fracture toughness, K_{IC} (MPa√m)
1700 °F/2 h–500 °F/2 h	
Before tempering	44.6 ± 0.8
After tempering	45.5 ± 1.1
1700 °F/2 h–600 °F/2 h	
Before tempering	64.25 ± 6
After tempering	40.0 ± 4.4
1700 °F/2 h–725 °F/2 h	
Before tempering	70.4 ± 4.4
After tempering	28.8 ± 4.6

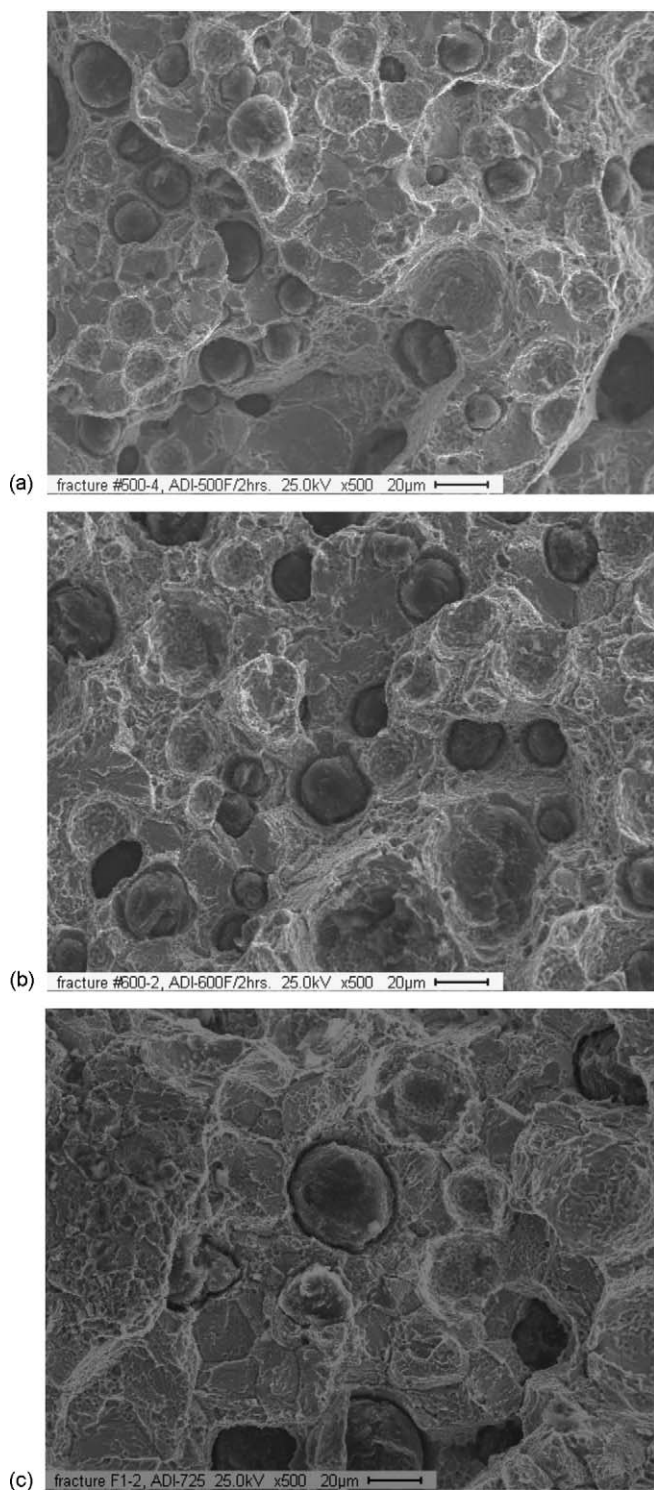


Fig. 11. Fractured surface at 500 $\times$ magnification: (a) heat treatment, 1700 °F/2 h–500 °F/2 h; (b) heat treatment, 1700 °F/2 h–600 °F/2; (c) heat treatment, 1700 °F/2 h–725 °F/2 h.

voids with a displacement of the graphite nodules as shown in Fig. 11(a). Hence, the low fracture toughness of the samples austempered at 500 °F is directly related to this fracture mode. The stress induced martensite formation in ADI at this temperature might be another reason for the brittle fracture [35,36] at this temperature. When carbon content of the austenite is low,

ADI is prone to formation of stress induced martensite and this causes reduction in fracture toughness. In the 500 °F austempered samples the carbon content of the austenite was lower i.e. austenitic carbon ($X_{\gamma}C_{\gamma}$) was lower. This has caused stress induced martensite formation in these samples and thus lower fracture toughness.

3.2.5.2. Tempered samples. Fig. 12(a–c) shows the fractographs of the tempered samples. It is observed from Fig. 12(c), that the fracture surface of 725 °F austempered samples after tempering consists of cleavage type of fracture. This totally brittle nature of the fracture surface is the reason for very low fracture toughness of this material. On the other hand there was not much difference in the appearance of the fracture surface at 500 °F tempered and non-tempered samples. So there was no difference in fracture toughness of these samples.

3.3. Thermal conductivity and magnetic properties

In order to measure the thermal and electrical transport properties of the different samples, we cut 2 mm $\times$ 2 mm $\times$ 8 mm long segments from each heat treated sample. We used conducting silver epoxy to attach gold-coated copper strips to each end of the bar, and to attach two additional strips along the length of the bar, evenly spaced from the middle. The thermal conductivity and electrical resistivity of the samples were measured simultaneously between $T = 10$ and 300 K using the thermal transport option on a Quantum Design PPMS. There is some uncertainty in determining the sample geometry, which will be propagated into the determination of the thermal and electrical transport properties of the samples. The thermal conductivities and electrical resistivity we measured have approximately 10% uncertainty, which is reflected on the y-axis values in Figs. 14 and 15.

The temperature dependent electrical resistivity for the six different samples is plotted in Fig. 14. As mentioned earlier, the geometrical factor used to calculate the resistivity has a rather large uncertainty, but this factor is fixed for each sample. The room temperature resistivity for all samples falls between 0.7 and 0.9 $\mu\Omega$. This range of values is similar to typical ADI material. Within the error arising from our estimate of the geometrical factor, there is no significant difference in room-temperature resistivity between the tempered and non-tempered samples. By investigating the temperature dependence of the resistivity, we can calculate the residual resistance ratio (RRR) for each sample, defined as the ratio of room temperature resistance to the resistance at $T = 4.2$ K. For all samples, the RRR value falls between 1.3 and 1.38. This suggests that structural defects remain the main source of electrical resistance at room temperature. It should be noted that the RRR values for the tempered and non-tempered samples are very similar, suggesting that tempering the samples does not significantly change the distribution of defects. This is consistent with the microstructural analysis, showing that the intrinsic defects remain unaffected by tempering process.

Motivated by suggestions that (ferritic) structures may provide better thermal properties [20,21], we have also measured the thermal conductivity of the tempered and non-tempered

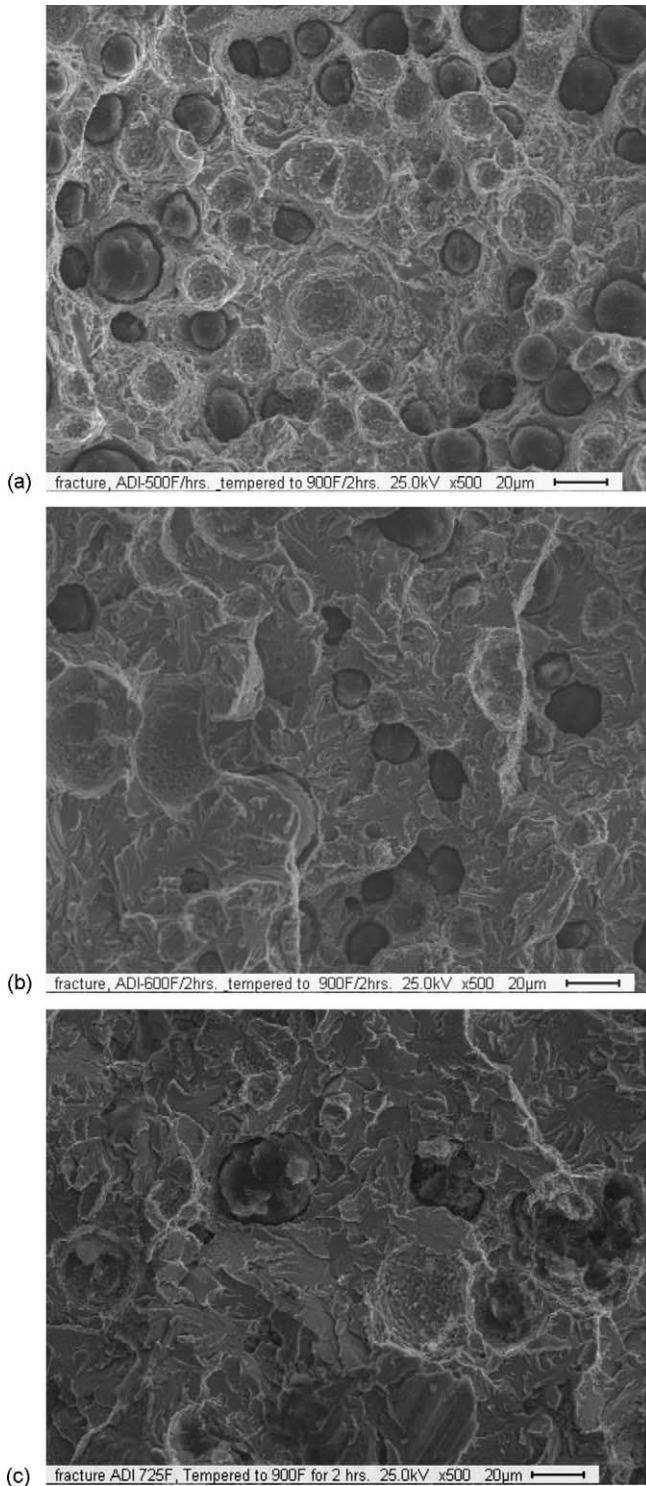


Fig. 12. Fractured surface at 500 $\times$ magnification: (a) heat treatment, 1700 $^{\circ}$ F/2h–500 $^{\circ}$ F/2h and tempered at 900 $^{\circ}$ F/2h; (b) heat treatment, 1700 $^{\circ}$ F/2h–600 $^{\circ}$ F/2h and tempered at 900 $^{\circ}$ F/2h; (c) heat treatment, 1700 $^{\circ}$ F/2h–725 $^{\circ}$ F/2h and tempered at 900 $^{\circ}$ F/2h.

samples. The thermal conductivities are plotted as a function of temperature in Fig. 15. The room temperature thermal conductivity for all samples is approximately 18 W/m K, with the exception of the sample austempered at 725 $^{\circ}$ F, which has a somewhat smaller thermal conductivity (13 W/m K). These val-

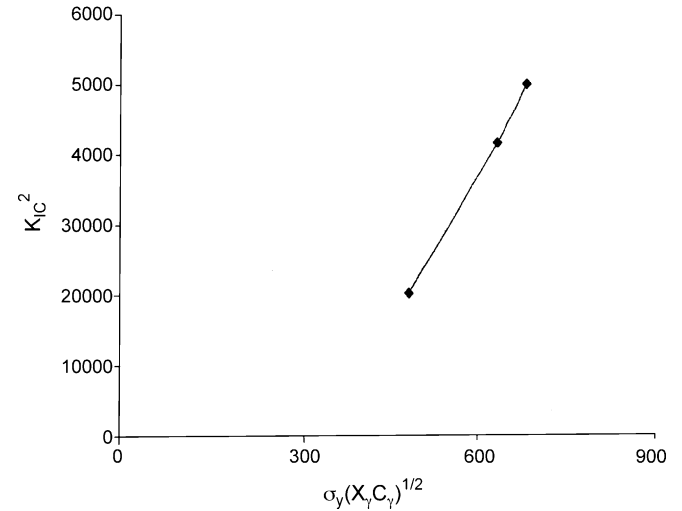


Fig. 13. A plot of (K_{IC}^2) against $\sigma_y(X_\gamma C_\gamma)^{1/2}$.

ues are again similar to the thermal conductivities of typical ADI. The temperature dependent thermal conductivity curves for all samples monotonically decrease with decreasing temperature. This suggests that the mean free path is already defect limited by room temperature, consistent with the low RRR values obtained. These measurements suggest that increasing the thermal conductivity to improve thermal shock and thermal fatigue properties will require reducing the number of defects scattering sites in these materials. Since the larger grain size of the tempered 725 $^{\circ}$ F austempered samples reduces the thermal conductivity; it may

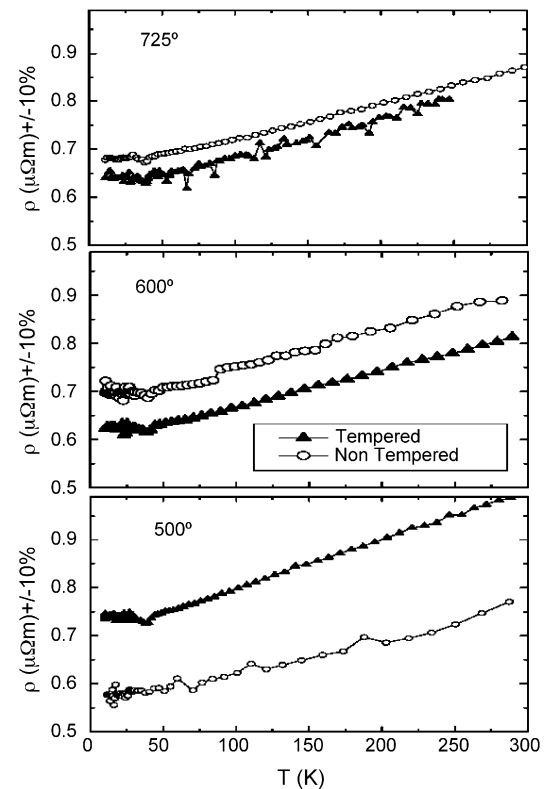


Fig. 14. Comparison of electrical resistivity.

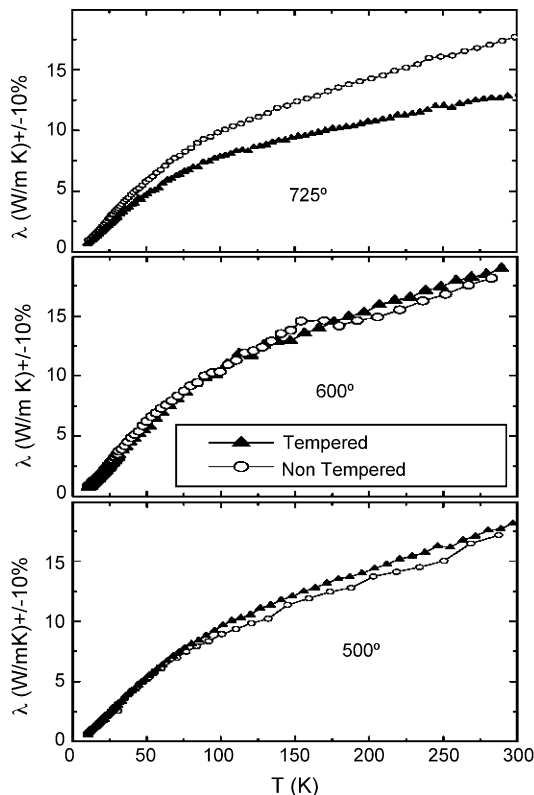


Fig. 15. Comparison of thermal conductivity.

be possible to increase the thermal conductivity by reducing the grain size. These measurements show that the thermal and electrical properties of ADI are dominated by defects and not by the lattice structure of the phases.

4. Conclusions

1. A fully ferritic structure (austenite free) ADI, was produced by tempering conventionally austempered samples at 484 °C (900 °F) for 2 h.
2. X-ray data confirmed that this ADI is fully ferritic (with no trace of austenite).
3. In the samples which were initially austempered at 260 °C (500 °F) and later on tempered at 484 °C (900 °F) for 2 h, there was not much reduction in mechanical properties and fracture toughness. This is correlated with the fine grain size of this material.
4. For the samples which were initially austempered at 385 °C (725 °F) and later on tempered at 484 °C (900 °F) for 2 h, there was drastic reduction in the ductility and fracture toughness.
5. Our preliminary investigation shows no significant changes in thermal and electrical properties on tempering of the conventionally austempered samples at 484 °C (900 °F). Because the mechanical properties are suitable (500 °F austempered samples) for wide range of applications we continue to investigate the methods to improve thermal properties.

6. This preliminary investigation shows that the thermal and electrical properties are dominated by the intrinsic defects and not by the crystal structure of the phases in ADI.
7. The main result of this investigation is that if we austemper ADI at 260 °C (500 °F) and temper at 484 °C (900 °F) for 2 h, it is possible to produce a fully ferritic structure in ADI without compromising its mechanical and thermal properties.

In all likely hood this will improve the machinability of ADI.

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